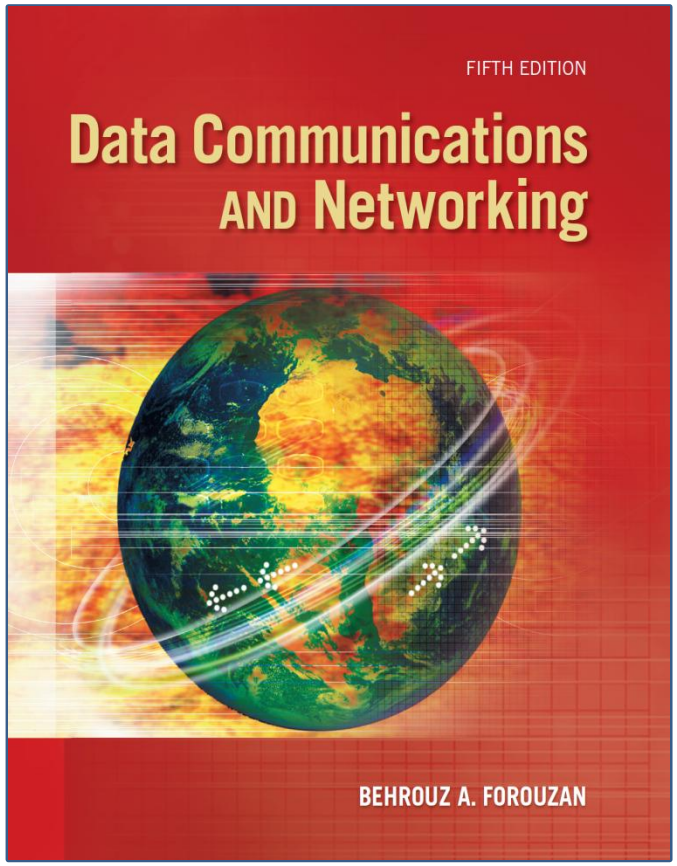


Network Services (Cloud Computing)

Lecturer: Dr. Khaled BinAbdl

Lecture 8: Cloud Computing

Cloud Computing

- **Cloud computing** is when a **third-party** service or vendor makes **computer resources available** to its customers **over the Internet**.

- **Cloud computing provides** many types of resources, including:
  - Applications
  - network services
  - Websites
  - database servers
  - specialized developer applications
  - or even entire operating systems.

Cloud Computing Elements

– All cloud computing service models incorporate the following elements:

– ***Service at any time***.
– ***Elastic services and storage***.
– ***Support for multiple client platforms.***
– ***Resource pooling and consolidation***.

Cloud Computing Elements

Service at any time.

– **On-demand** service is available to users at any time.

– **Cloud computing** vendors often advertise uptime of their services, which is the percentage of time in any given year when their services were available online without disruption.

Cloud Computing Elements

Elastic services and storage.

- **Rapid elasticity** refers to the service's ability to be scaled up or down as the need level changes for a particular customer without requiring hardware changes that could be costly for the customer.

- Layers of services, such as applications, storage space, or number of users, can be added or removed upon request or automatically, depending upon the options made available by the service vendor.

Cloud Computing Elements

Support for multiple client platforms.

- **Platform** refers to the operating system, the runtime libraries or modules the OS provides to applications, and the hardware on which the OS runs**.**

- Cloud resources are made available to clients through standardized access methods that can be used with a variety of platforms, such as Windows, Linux, or Mac OS X, on any number of devices, such as desktops, laptops, tablets, and smart phones from various manufacturers.

Cloud Computing Elements

Resource pooling and consolidation.

– With **resource pooling**, services to multiple customers are hosted on shared physical resources, which are dynamically allocated to meet customer demand.

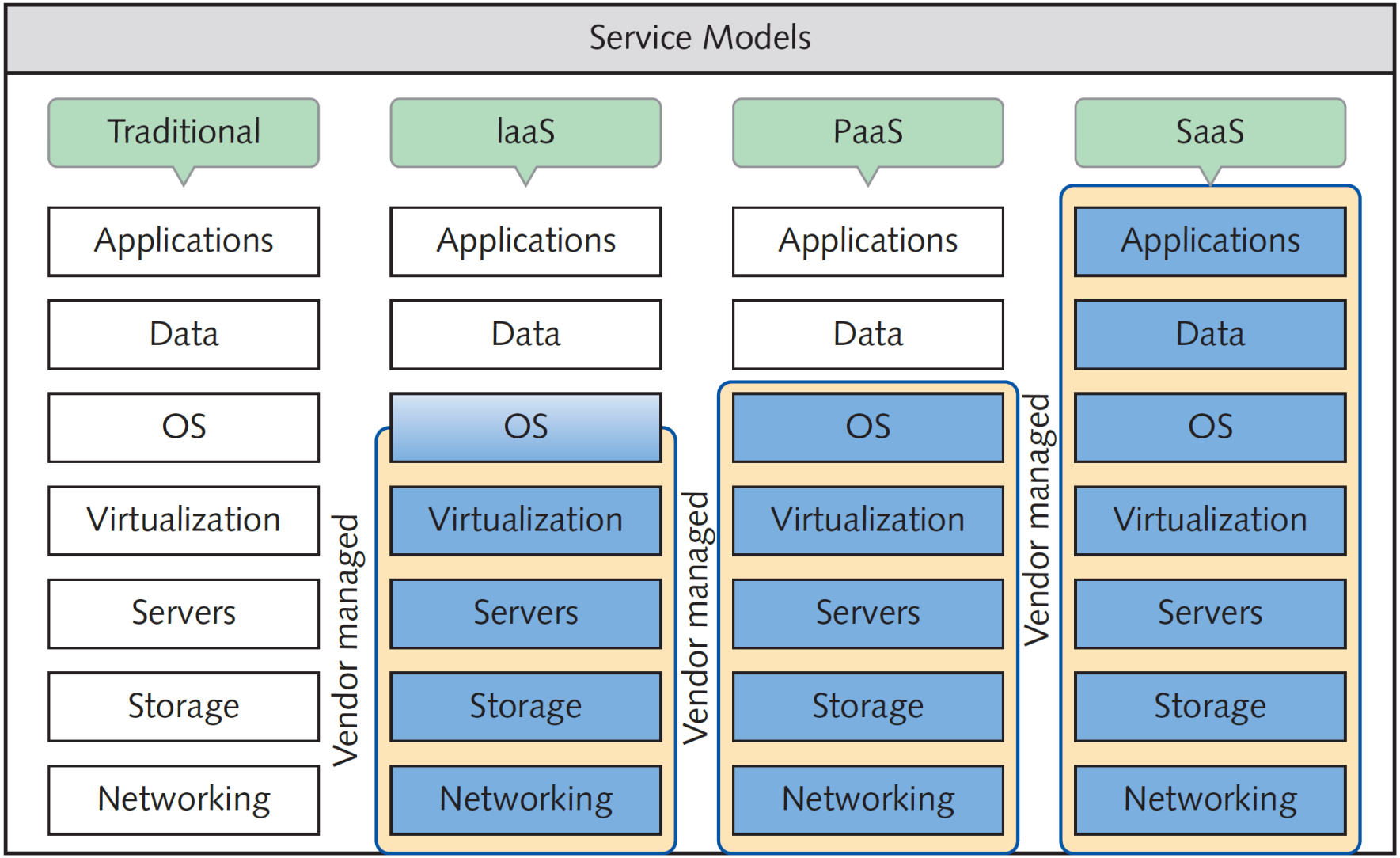
– Customers generally are not aware of where the physical devices providing cloud services are located geographically, which is called **location independence**.

Cloud Computing Elements

Measured service.

- Resources offered by a cloud computing vendor, such as storage, applications, bandwidth, and other services, are measured, or metered, for billing purposes and/or for the purpose of limiting any particular customer's use of that resource according to the service agreement.

- These **measured services** have reporting policies in place to ensure transparency between vendors and customers.

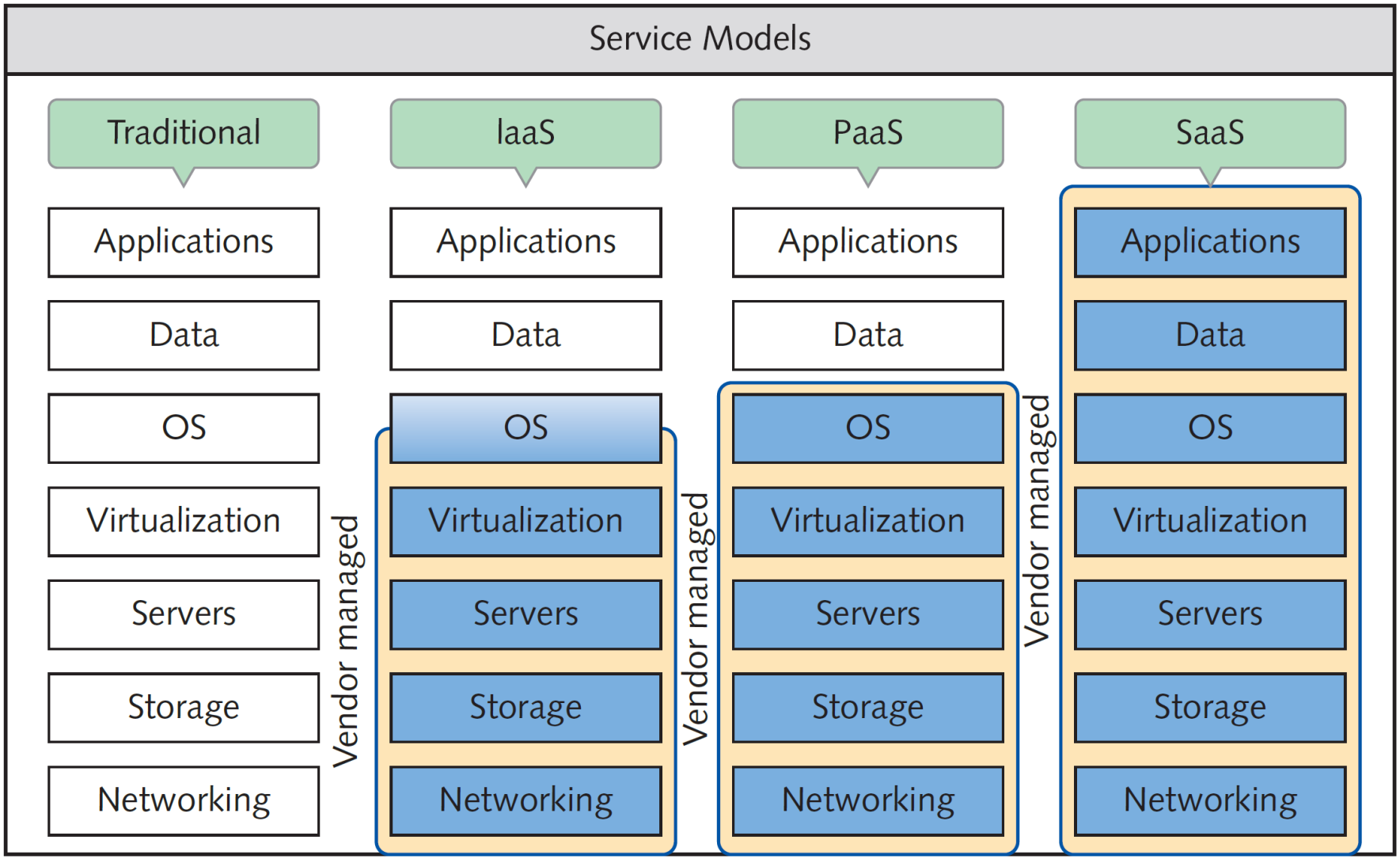

Service Models
Traditional
IaaS
PaaS
SaaS
Applications
Data
OS
Virtualization
Servers
Storage
Networking
Vendor managed
Vendor managed
Vendor managed

Cloud Computing Categories

- Cloud computing service models are categorized by the types of services they provide.

- National Institute of Standards and Technology (NIST) has developed a standard definition for each category, which varies by the division of labor implemented.

Cloud Computing Categories



Cloud Computing Categories

IaaS (Infrastructure as a Service).

– With **IaaS (Infrastructure as a Service)**, the customer rents hardware, including servers, storage, and networking, and can use these hardware services virtually.

– Customers are responsible for their own application installations, data management, and backup. In most situations, customers are also responsible for their own operating systems.

Cloud Computing Categories

– ***PaaS (Platform as a Service).***

– With **PaaS (Platform as a Service)**, a customer rents hardware, operating systems, and some applications that might support other applications the customer may install.

– PaaS is popular with software developers who require access to multiple platforms during the development process.

Cloud Computing Categories

SaaS (Software as a Service).

- With **SaaS (Software as a Service)**, customers rent hardware, operating systems, and applications specific to the customer's needs.
- Applications are provided through an online user interface and are compatible with a multitude of devices and operating systems.
- Online email services, such as Gmail and Yahoo!, are good examples of SaaS.

Cloud Computing Categories

XaaS (Anything as a Service or Everything as a Service).

- In the **XaaS (Anything as a Service** or **Everything as a Service)** model, the “X” represents an unknown, just as it does in algebra.

- Here, the cloud can provide any combination of functions depending on a customer’s exact needs.

Deployment Models for Cloud Computing

Public cloud.

– In a **public cloud**, services are provided over the Internet to the general public.

– Google or Yahoo! email services are examples of public cloud deployment.

Deployment Models for Cloud Computing

Private cloud.

– In a **private cloud**, services are established on an organization's own servers, or established virtually for a single organization's private use.

– For **example**, an insurance company might have a centralized data center that provides cloud services in a private cloud to branch offices throughout the United States.

Deployment Models for Cloud Computing

Community cloud.

- In a **community cloud**, services are shared between multiple organizations, but not available publicly.
- Organizations with common interests, such as government requirements, might share resources in this way.
- For **example**, a medical database might be made accessible to all hospitals in a geographic area. In that case, the community cloud could be hosted internally by one or more of the organizations involved, or hosted by a third-party provider.

Deployment Models for Cloud Computing

Hybrid cloud.

– A **hybrid cloud** is a combination of public, private, and community clouds used by the same organization.

– For **example**, a company might store data in a private cloud, but use a public cloud email service.

Discussion